Name :- Kartik Ram Birajdar

Roll No:- A-13

Branch:- AI&DS

Assignment:-08

Aim:-

To implement and analyze correlation and regression techniques using Python, including: Karl Pearson's Correlation Coefficient Simple Linear Regression And to understand the relationship between variables and perform prediction.

1. **What is Correlation? How is it different from Regression?**

Correlation measures the strength and direction of relationship between two variables. Value ranges from –1 to +1 +1 → Perfect positive relationship –1 → Perfect negative relationship 0 → No relationship

Basis	Correlation	Regression
Purpose	Measures relationship	Predicts values
Direction	No cause-effect	Shows dependence
Variables	Symmetrical	One dependent, one independent
Output	Coefficient (r)	Equation (y = mx + c)

2. **What is Simple Linear Regression?**

(Example) Simple Linear Regression models the relationship between two variables using a straight line: y=mx+c Where: y = dependent variable x = independent variable m = slope c = intercept Example: Suppose: x = Hours studied y = Marks scored If regression equation is: y=5x+10 Then: If a student studies 4 hours → y = 5(4) + 10 = 30 marks

3. How do you calculate Pearson Correlation in Python? You can use NumPy or Pandas. Using NumPy:

```
ipmrpionrtt(" Pneuamrpsyo na sC onrpelation:", correlation)

x = [10, 20, 30, 40, 50]
y = [15, 25, 35, 45, 55]

correlation = np.corrcoef(x, y)[0, 1]
print("Pearson Correlation:", correlation)
```

```
Pearson Correlation: 1.0
```

```
import pandas as pd

data = pd.DataFrame({
    'x': [10, 20, 30, 40, 50],
    'y': [15, 25, 35, 45, 55]
})

print(data.corr())
```

```
     x    y
x  1.0  1.0
y  1.0  1.0
```

4. What is Multicollinearity? Does it affect Simple Regression? Multicollinearity occurs when independent variables are highly correlated with each other.

Example: x1 = Study hours x2 = Number of classes attended (Both may be related) Effect: Causes unstable coefficients Makes interpretation difficult Important Point: 👉 It does NOT affect simple regression, because: Simple regression has only one independent variable Multicollinearity occurs only in multiple regression

5. Strong Correlation but Poor Prediction Accuracy — Why?

This situation can happen due to several reasons:

1. Non-linear Relationship Correlation measures linear relationship only If data is curved → prediction fails
2. Outliers Extreme values distort regression line
3. Overfitting / Underfitting Model doesn't generalize well
4. High Variability Data points are scattered despite trend
5. Correlation ≠ Causation Two variables may move together but not truly influence each other

2. Simple Linear Regression

```
from sklearn.linear_model import LinearRegression
import numpy as np
```

```
# Sample data
x = np.array([10, 20, 30, 40, 50]).reshape(-1, 1)
y = np.array([15, 25, 35, 45, 60])

# Create model
model = LinearRegression()
model.fit(x, y)

# Predict value
predicted = model.predict([[35]])

print("Slope (m):", model.coef_[0])
print("Intercept (c):", model.intercept_)
print("Predicted value for x=35:", predicted[0])
```

```
Slope (m): 1.1
Intercept (c): 3.0
Predicted value for x=35: 41.5
```

3. Plotting Regression Line

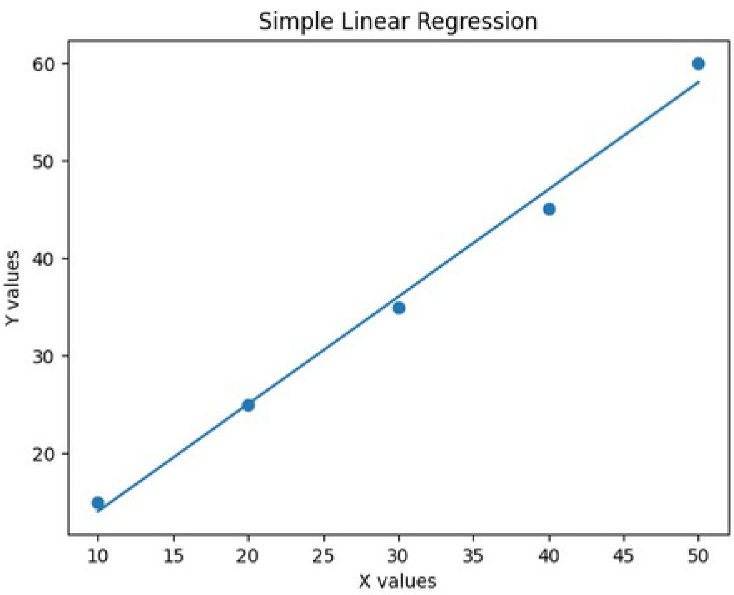
```
import matplotlib.pyplot as plt

# Plot data points
plt.scatter(x, y)

# Plot regression line
plt.plot(x, model.predict(x))

plt.xlabel("X                    values")
plt.ylabel("Y                    values")
plt.title("Simple Linear Regression")

plt.show()
```



Conclusion

Correlation helps understand relationship strength Regression helps in prediction Pearson’s method is widely used for measuring linear correlation Simple regression is easy but limited Strong correlation does not guarantee accurate prediction